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Measuring preferences for algorithms — How willing are people to cede control to algorithms?☆

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ABSTRACT

We suggest a simple method to elicit individual preferences for algorithms. By altering the monetary incentives for ceding control to the algorithm, the menu-based approach allows for measuring in particular the degree of algorithm aversion. Using an experiment, we elicit preferences for algorithms in an environment with measurable performance accuracy under two conditions — the absence and the presence of information about the algorithm's performance. Providing such information raises subjects' willingness to rely on algorithms when ceding control to the algorithm is more costly than trusting in their own assessment. However, algorithms are still underutilized.

1. Introduction

Algorithms are omnipresent and have become a fundamental part of our daily life. When solving problems, humans are increasingly faced with the choice of whether to delegate decisions to algorithms in business, medicine, public policy, as well as in everyday situations. Two phenomena dominate the ongoing discussion on the interaction between humans and algorithms: algorithm aversion and algorithm appreciation. Algorithm aversion describes the human reluctance to rely on decisions or recommendations made by an algorithm in favor of relying on the own judgment (Dietvorst et al., 2015). Algorithm appreciation is defined as the opposite phenomenon: a human decision-maker prefers algorithmic judgment to human judgment (Logg et al., 2019).[1]

The literature has identified a number of factors that influence the presence of algorithm aversion, *e.g.*, performance feedback (Alexander et al., 2018; Dietvorst et al., 2015; Jung & Seiter, 2021), the characteristics of the task (Dietvorst & Bharti, 2020), response time of the algorithm (Efendić et al., 2020), time pressure (Jung & Seiter, 2021), learning (Filiz et al., 2021), the illusion of better understanding human than algorithm decision-making (Bonezzi et al., 2022), or consequences of failure (Filiz et al., 2023). Several factors influencing algorithm appreciation have also been identified, such as the nature of the task (Logg et al., 2019), performance feedback (You et al., 2022), and incentives (Greiner et al., 2022), to name a few. An increasing number of papers focus on finding ways to overcome algorithm aversion and increase algorithm appreciation. This includes permitting subjects to modify the decisions produced by an algorithm or to select which information the algorithm processes (Dietvorst et al., 2018; Jung & Seiter, 2021; Sele & Chugunova, 2024), as well as demonstrating the algorithm's ability to learn and improve (Berger et al., 2021; Reich et al., 2023).[2]

Despite the large body of literature on algorithm aversion and algorithm appreciation, far too little is known about the strength of the preferences for algorithms, in particular the degree of algorithm aversion.[3] The methodology used so far involves the choice between

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[1] For a comprehensive interdisciplinary review on how humans interact with automated agents, see Chugunova and Sele (2022).

[2] Jussupow et al. (2020) provide an overview of empirical evidence on algorithm aversion and appreciation. Mahmud et al. (2022) review the factors that affect algorithm aversion. Kaufmann et al. (2023) review the task-dependence of algorithm advice acceptance.

[3] One approach to measure preferences for algorithms is by eliciting the Weight on Advice (WOA) via the Judge Advisor System (JAS) used, *e.g.*, in Logg et al. (2019). In the JAS, participants may revise an initial judgment after receiving advice. Participants' algorithm use is then measured as the WOA. The WOA ranges from 0% to 100% when a participant revises their judgment toward the advice. WOA is 0% when the advice is ignored or the revised judgment is in the opposite direction of the advice. Thus, the JAS allows for a conclusive measurement only of the degree of algorithm appreciation.

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one's own decision and the algorithm's decision when facing a problem on what might be called a ceteris paribus basis. That is, with all things being equal, which judgment would subjects prefer? This approach neither captures the heterogeneity of preferences for algorithms nor takes the costs of employing algorithms into account. While it allows for measuring the effects of interventions aiming to overcome algorithm aversion at the extensive margin, it misses the effects at the intensive margin. In this paper, we take the investigation a step further and suggest a simple method that allows for the measurement of the intensity of the preferences for algorithms.

Consider a payoff-relevant task that can be performed by an algorithm. The method allows for finding an answer to the question: "How many monetary benefits are subjects willing to give up to avoid the algorithm?" It involves presenting subjects with a menu of simple choice problems. In each problem, subjects can choose between performing the task on their own or delegating it to the algorithm. The problems differ with respect to the performance-based payoff when the algorithm is chosen. By varying the payoff for employing the algorithm, the menu-based approach allows for measuring the degree of algorithm aversion.

We apply the suggested fine-grained method and elicit subjects' algorithm attitudes under two conditions—the absence and the presence of information about the algorithm's performance. Our experimental results show the existence of potentially strong preferences for algorithms. In both conditions, subjects exhibit lower algorithm aversion compared to the incentive-based prediction, *i.e.*, when taking into account only the size of the payment reduction for inaccurate forecasts. Still, subjects underutilize algorithms compared to a performance-based prediction, *i.e.*, when taking into account feedback on the subject's own performance and the algorithm's performance. In the presence of feedback on the algorithm's performance, we find a correlation of predicted and observed algorithm choices. This correlation is largely driven by beliefs about the algorithm's and the own performance. Providing information about the algorithm's performance positively affects the use of algorithms.

When applying the binary measure commonly used in previous studies to our data, we are unable to detect either the positive effect on the algorithm's performance or the role of beliefs. This demonstrates that the suggested menu-based approach can uncover differences in algorithm preferences that methods which rely on ceding control to algorithms, when the payoff consequences are identical, might miss.

2. Experimental design

We apply the suggested method to elicit preferences for algorithms in a forecasting task used in several experimental studies, which gives subjects a choice between making a forecast decision themselves or delegating it to an algorithm (*e.g.*, Dietvorst et al., 2015; Jung & Seiter, 2021; Logg et al., 2019). Those studies typically report that participants prefer the algorithm's decision over their own in 40 to 60% of cases, providing sufficient margins for both algorithm appreciation and aversion. Moreover, performance in forecasting tasks can be precisely measured after observing the realized outcomes. Furthermore, these tasks enable the elicitation of algorithm performance on the same scale as human performance.

As performance feedback plays a crucial role in the occurrence of algorithm aversion (see surveys by Jussupow et al., 2020; Mahmud et al., 2022, and the studies cited therein), we used a between-subjects design and implemented two conditions, which differed with respect to the feedback participants received. In the "Human"-condition, participants only got feedback on their performance. In the "Human-Algorithm"-condition, they saw the algorithm's performance in addition to their own performance. By keeping the experience with the task constant (*i.e.*, providing feedback on participants' performance) between conditions, we can isolate and measure the effect of feedback on the algorithm's performance.

Following Dietvorst et al. (2015), Logg et al. (2019), and Jung and Seiter (2021), our experiment consisted of 11 rounds. In the first 10 unincentivized rounds, participants were asked to forecast the rank of a randomly chosen U.S. state in terms of departing air passengers in 2011 based on five pieces of information (number of major airports, rank of population, rank of county, rank of median household income, rank of expenditure on domestic travel). In both conditions, participants got feedback comprising this information, their forecast, and the true rank. In the Human-Algorithm condition, they additionally received information on what rank an algorithm predicted for this task.[4]

In the final payoff-relevant round, subjects first had to decide whether their own forecast or the forecast of the algorithm should be used to determine their payoff in that round. For each unit of deviation of the forecast from the true rank, the payoff was reduced by the penalty rate X. More precisely, the payment rule was

$$\text{payoff} = 7 - X * |\text{forecast} - \text{true rank}|$$

in EUR. Thus, the unit of deviation in a forecast is the measure for forecasting inaccuracy.

In contrast to the previous studies, where subjects made a single choice between using their own forecast or the algorithm's forecast to determine their payoff, participants in our experiment made decisions in nine choice problems, see Table 1. For the case in which the subject's own forecast becomes payoff-relevant, the penalty rate X_{own} remains constant across all nine problems. For the case that the algorithm's forecast is used to determine the subject's payoff, the penalty rate X_{alg} increases when going down the list of choice problems. In the first choice problem, the algorithm forecast is payoff-irrelevant ($X_{\text{alg}} = 0$). In the fifth choice problem, the penalty rate is the same regardless of the type of forecast, human or algorithm ($X_{\text{own}} = X_{\text{alg}} = 0.12$). Hence, the fifth choice problem corresponds to the single choice subjects usually faced in previous studies (*e.g.*, Dietvorst et al., 2015; Jung & Seiter, 2021; Logg et al., 2019).

A subject who is only concerned with the size of the payment reduction would stop choosing the algorithm when $X_{\text{alg}} > X_{\text{own}}$ (incentive-based prediction).

Taking performance feedback into account, it would be optimal to stop choosing the algorithm for

$$X_{\text{alg}} > X_{\text{own}} * \frac{|\text{own forecast} - \text{true rank}|}{|\text{algorithm's forecast} - \text{true rank}|},$$

where |own forecast – true rank| and |algorithm's forecast – true rank| are the individual averages of the performances in the 10 unincentivized rounds (performance-based prediction).

After subjects decided on all nine choice problems, they had to make a forecast once again. At the end of the round, subjects were asked questions that dealt with their confidence regarding their own and the algorithm's forecasting performance. Furthermore, they were asked about their beliefs regarding the expected error in their own forecast and in the algorithm's forecast (see Table A.1 in the Appendix for the set of questions).

At the end of the experiment, one of the nine choice problems was randomly selected and implemented for real payment. Participants had to fill in a post-experimental questionnaire that contained, among other things, a self-assessed, hypothetical measure of risk attitude on a 0–10 scale (SOEP). We also elicit subjects' attitudes toward ambiguity. Altogether, the experiment encompassed a total of 212 subjects (106 per condition). The experiment was conducted with students at the Technische Universität Berlin (47%/52% female in the Human/Human-Algorithm condition). Participants were invited to the experiment with ORSEE (Greiner, 2015). All experimental sessions were conducted using z-Tree (Fischbacher, 2007). The average length of a session was 45 min. The average total earnings per participant was 13.68 EUR, including a show-up fee of 6 EUR.[5]

[4] We used the original algorithm employed in Experiment 3A/B by Dietvorst et al. (2015).

[5] See Web Appendix for a translated version of the instructions.

Table 1
The payoff structure of the nine paired choice problems.

Choice problem	Payoff (own forecast) in EUR $7 - X_{own} * \|\text{forecast} - \text{true rank}\|$	Payoff (algorithm forecast) in EUR $7 - X_{alg} * \|\text{forecast} - \text{true rank}\|$	Penalty rate difference $(X_{alg} - X_{own})$
1	$7 - 0.12 * \|$ forecast − true rank$\|$	$7 - 0 * \|$ forecast − true rank$\|$	−0.12
2	$7 - 0.12 * \|$ forecast − true rank$\|$	$7 - 0.03 * \|$ forecast − true rank$\|$	−0.09
3	$7 - 0.12 * \|$ forecast − true rank$\|$	$7 - 0.06 * \|$ forecast − true rank$\|$	−0.06
4	$7 - 0.12 * \|$ forecast − true rank$\|$	$7 - 0.09 * \|$ forecast − true rank$\|$	−0.03
5	$7 - 0.12 * \|$ forecast − true rank$\|$	$7 - 0.12 * \|$ forecast − true rank$\|$	0
6	$7 - 0.12 * \|$ forecast − true rank$\|$	$7 - 0.15 * \|$ forecast − true rank$\|$	0.03
7	$7 - 0.12 * \|$ forecast − true rank$\|$	$7 - 0.18 * \|$ forecast − true rank$\|$	0.06
8	$7 - 0.12 * \|$ forecast − true rank$\|$	$7 - 0.21 * \|$ forecast − true rank$\|$	0.09
9	$7 - 0.12 * \|$ forecast − true rank$\|$	$7 - 0.24 * \|$ forecast − true rank$\|$	0.12

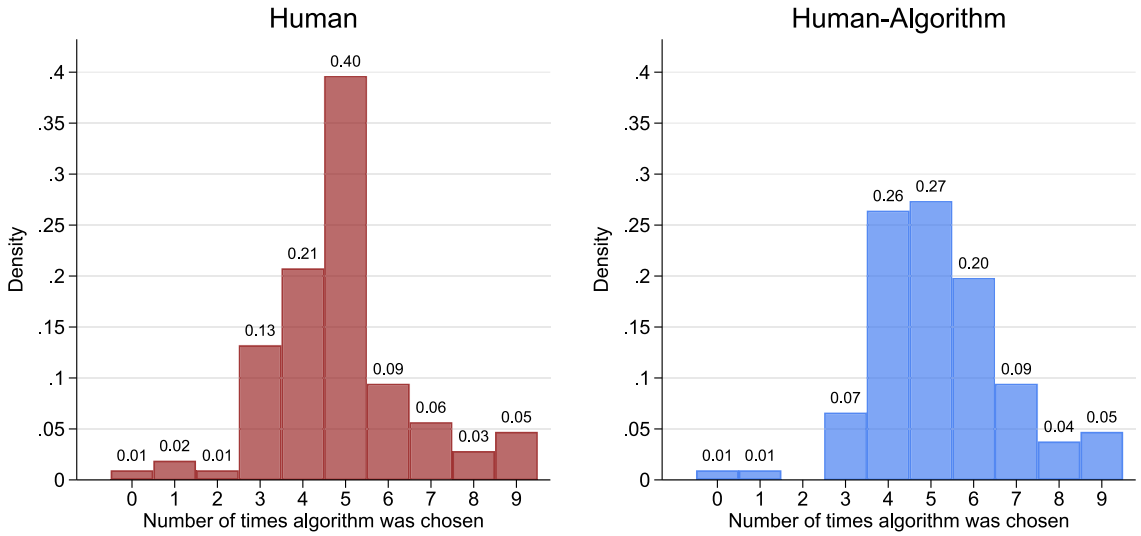


Fig. 1. Relative share of the number of times the algorithm was chosen per condition.

3. Results

Fig. 1 illustrates the heterogeneity of participants' preferences for algorithms by showing the number of times participants chose the algorithm for each condition. The number of algorithm choices is an indicator of the degree of algorithm aversion. In Human, the number of times the algorithm was chosen peaks at five, while it peaks at both four and five in Human-Algorithm. In Human, 17% of the participants choose the algorithm zero to three times, opposed to 9% in Human-Algorithm. In contrast, in Human-Algorithm 38% of the participants choose the algorithm six to nine times, opposed to 23% in Human. The majority of participants (94% in each condition) are neither completely reliant on the algorithm nor do they fully cede control to the algorithm.

Next, we check whether the participants' choices are consistent and purposeful. In the first choice problem, where the penalty rate is zero for the algorithm's forecast, 95%, respectively 93%, of the participants choose the algorithm in Human, respectively Human-Algorithm. We consider participants as having monotonic preferences when they switch at most once from choosing the algorithm to choosing oneself. 85% of the participants in Human and 69% of the participants in Human-Algorithm exhibit monotonic preferences. Note that for monotonic preferences the number of algorithm choices designates a unique crossover point from choosing the algorithm to choosing oneself.[6]

Fig. 2 presents the proportion of algorithm choices for each of the nine choice problems per condition for all participants (left graph) and those who exhibit monotonic preferences (right graph). The figure includes reference lines representing the incentive-based prediction (dotted line) and the performance-based prediction (dashed line) based on data from both conditions. For the derivation of both predictions, see Section 2. In Human, participants behave close to the incentive-based prediction: the share of algorithm choices drops from 81% in problem 4 to 22% in problem 6 (83 to 22% in the case of monotonic preferences). In Human-Algorithm, the proportion of algorithm choices is close to both (aligned) predictions in problems 1 to 4, particularly in the case of monotonic preferences. In problems 6 to 9, where both predictions are not aligned, we observe a higher usage of the algorithm compared to the incentive-based prediction.

Compared to the performance-based prediction participants underutilize the algorithm in both conditions. Yet, we find a positive correlation between the observed and the optimal number of times

[6] See Fig. A.1 in the Appendix for the distributions of the number of switches. The mode and median number of switches are one in both conditions. 85% of the participants in Human and 72% of the participants in Human-Algorithm switch at most once. A higher number of switches might indicate uncertainty about the point of indifference. Instead of using the number of algorithm choices, an alternative way to deal with such non-monotonic preferences is to consider intervals of penalty rates. The lower (upper) bound of an interval is determined by the first (last) switching point. Results do not change when using those intervals as the dependent variable in the regression analysis.

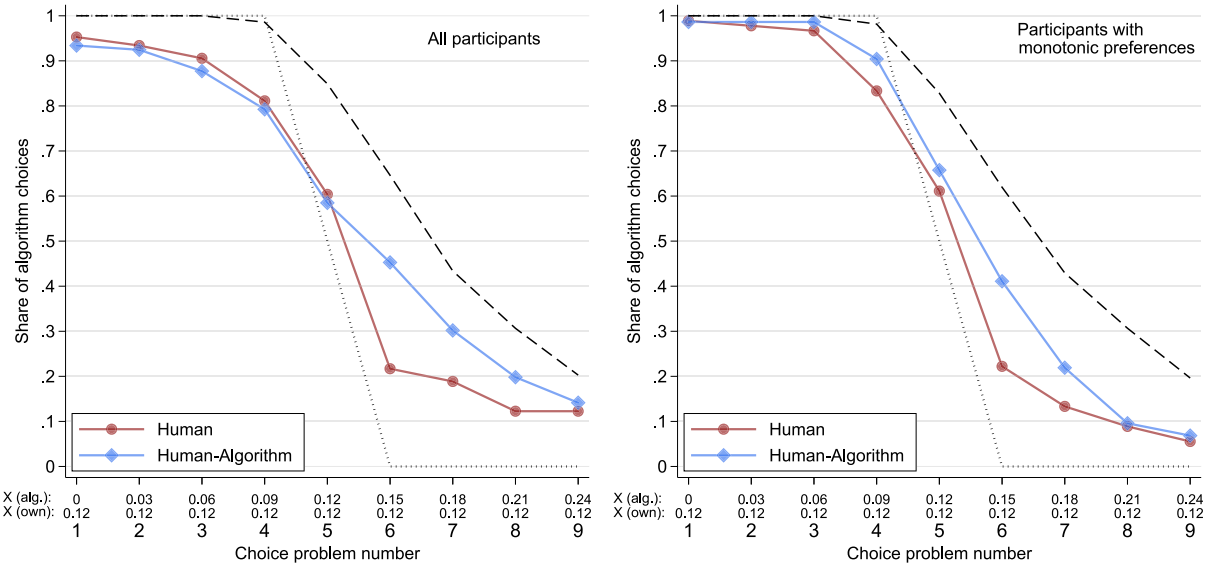


Fig. 2. Share of algorithm choices by choice problem number (and corresponding X). The incentive-based prediction (dotted line) and the performance-based prediction (dashed line) are computed using data from both conditions.

the algorithm was chosen in Human-Algorithm (Spearman's rank correlation, all participants: $r(105) = 0.4056$, $p < 0.01$, participants with monotonic preferences: $r(72) = 0.4414$, $p < 0.01$). This is not the case in Human, where participants do not receive feedback on the algorithm's performance (all participants: $r(105) = 0.1154$, $p = 0.24$, participants with monotonic preferences: $r(89) = 0.1349$, $p = 0.20$).[7]

In both conditions, participants behave very similarly up to choice problem 5 ($X_{own} = X_{alg}$). This changes substantially in choice problems 6 to 9, where $X_{own} < X_{alg}$. The proportion of algorithm choices in Human-Algorithm is larger compared to Human, indicating that algorithm aversion decreases in the presence of feedback on the algorithm's performance. Indeed, based on a Wilcoxon rank-sum test, we find significant differences between the number of algorithm choices in Human ($M = 4.86$, $SD = 1.64$) and in Human-Algorithm ($M = 5.21$, $SD = 1.60$), with $p = 0.09$.[8]

Table 2 presents the OLS regression results on algorithm choices within conditions (specifications 1 to 4) and between conditions (specifications 5 and 6). In the even-numbered specifications, we consider only participants with monotonic preferences. In all specifications, we utilize belief and confidence measures while controlling for gender, risk, and ambiguity attitudes. As shown in specifications (1) and (2), the confidence in both the own and the algorithm's forecast explain the algorithm choices in the Human condition. An increased (decreased) confidence in the algorithm's (own) performance by one step on the 5-point Likert scale increases (decreases) the number of algorithm choices by around 0.5. The OLS regression results in (3) confirm the positive correlation between the observed and predicted number of algorithm choices in the Human-Algorithm condition. In the case of monotonic preferences, as apparent in (4), this correlation seems to be driven by beliefs about the own and the algorithm's accuracy of forecasts. The larger the belief about the own (algorithm's) forecasting error, the larger (lower) the number of algorithm choices. In (5) and (6), the effect of the presence of information on the algorithm's performance is captured by the "Human-Algorithm" dummy variable. Providing information about the algorithm's performance increases the number of algorithm choices by 0.5 on average.[9]

To assess the degree of algorithm aversion in terms of the methodology used so far in previous studies, we reduce our data to a binary form in two distinct ways. First, we classify participants with less than five algorithm choices as not choosing the algorithm, and participants with at least five algorithm choices as choosing the algorithm. The shares of participants who chose the algorithm obtained in this way are 62% in Human and 65% in Human-Algorithm. The difference between the two conditions is not statistically significant (two-sided Fisher's exact test, $p = 0.78$).[10] Second, we limit our data only to the single choice, where the penalty rate of the own and the algorithm's forecasting errors is equal (choice problem 5). This corresponds to the choices subjects usually faced in previous studies (*e.g.*, Dietvorst et al., 2015; Jung & Seiter, 2021; Logg et al., 2019). The shares of algorithm choices obtained in this way are 60% in Human and 58% in Human-Algorithm.[11] Again, the difference is not statistically significant (two-sided Fisher's exact test, $p = 0.89$). Using the binary measures as dependent variables in our regressions, we cannot confirm the results obtained with our fine-grained measure (see Table A.4 in Appendix A). Providing feedback on the algorithm's performance does not affect the tendency to choose the algorithm. Moreover, we are not able to detect the role of confidence in the own forecast in the Human condition and the role of beliefs in the Human-Algorithm condition.

This exercise shows that the suggested menu-based approach is able to reveal differences in algorithm preferences, where the commonly used binary measures might fail.

4. Conclusion

We suggest a simple method for the fine-grained elicitation of human preferences for algorithms. We apply it to measure how willing

[7] For the corresponding plots, see Fig. A.2 in Appendix.

[8] $p = 0.07$, Human (M=4.88, SD=1.67), Human-Algorithm ($M = 5.32$, $SD = 1.70$) in the case of monotonic preferences.

[9] Results remain qualitatively unchanged when (i) employing interval regression (Andersen et al., 2006), see Table A.3 in the Appendix, and (ii) controlling for learning, *i.e.*, by only using the second half of the 10 unincentivized rounds to determine the performance-based prediction.

[10] See Fig. 1 for the number of times the algorithm was chosen per condition.

[11] Note that previous studies report that subjects choose the algorithm in 40 to 60% of the cases.

Table 2
OLS regressions within and between conditions.

	Within conditions				Between conditions	
	Human		Human-Algorithm		Reference category: Human	
	(1)	(2)	(3)	(4)	(5)	(6)
Performance-based prediction	0.003	0.019	0.194**	0.064	0.091	0.050
	(0.076)	(0.082)	(0.080)	(0.111)	(0.055)	(0.063)
Human-Algorithm					0.474**	0.516**
					(0.188)	(0.209)
Confidence in						
forecast (own)	−0.476**	−0.538**	−0.266	−0.365	−0.395**	−0.467**
	(0.208)	(0.243)	(0.246)	(0.286)	(0.155)	(0.188)
forecast (alg.)	0.556***	0.673***	0.416	0.469	0.519***	0.601***
	(0.195)	(0.217)	(0.265)	(0.353)	(0.156)	(0.185)
Belief about						
deviation from true rank (own)	0.011	0.006	0.016	0.178**	0.011	0.036
	(0.034)	(0.037)	(0.019)	(0.081)	(0.015)	(0.037)
deviation from true rank (alg.)	0.004	0.020	−0.016	−0.209**	0.006	−0.006
	(0.042)	(0.053)	(0.033)	(0.104)	(0.025)	(0.048)
perfect prediction (own)	−0.394*	−0.400	−0.305*	−0.140	−0.379***	−0.396**
	(0.214)	(0.242)	(0.179)	(0.207)	(0.136)	(0.158)
perfect prediction (alg.)	0.389*	0.318	0.477***	0.399**	0.473***	0.491***
	(0.233)	(0.286)	(0.161)	(0.172)	(0.129)	(0.149)
Constant	4.522***	4.426***	2.720*	2.679*	2.935***	2.506**
	(1.255)	(1.440)	(1.439)	(1.358)	(1.013)	(1.030)
Gender	Yes	Yes	Yes	Yes	Yes	Yes
Ambiguity and risk aversion	Yes	Yes	Yes	Yes	Yes	Yes
Only monotonic	No	Yes	No	Yes	No	Yes
Observations	106	90	106	73	212	163
R^2	0.380	0.422	0.389	0.526	0.373	0.451

Standard errors in parentheses. The dependent variable is the number of algorithm choices.
* Significance at the 10% level.
** Significance at the 5% level.
*** Significance at the 1% level.

people are to cede control to algorithms under two conditions, the absence and the presence of information on the algorithm's performance.

We find that providing information about the performance of the algorithm reduces human reluctance to rely on algorithms. This effect would have been missed when applying the commonly used binary method. Our results reveal that the latter may be insufficient given the heterogeneity of human preferences and the complexity of the underlying decision problem.

The proposed method allows for inspecting the consistency of choices, distinguishing between monotonic and non-monotonic preferences, and capturing the variety of the degree of algorithm aversion among individuals. This heterogeneity could be an explanatory factor for economically consequential decisions on the use of algorithms in several domains, *e.g.*, delegation of hiring decisions to algorithms (Dargnies et al., 2024), relying on algorithm advice in price estimation tasks (Greiner et al., 2022), seeking advice of algorithms in moral dilemmas (Leib et al., 2024), or delegation to algorithms in financial decision-making (Germann & Merkle, 2023).

Measuring the strength of people's preferences for algorithms is the first step toward creating a standardized assessment of algorithm aversion. A standardized measurement can confirm existing research findings and facilitate the development and validation of simple survey items to evaluate algorithm aversion. Furthermore, the suggested method can be employed to distinguish preferences for algorithms from preferences for external advice in general. For example, one could measure the willingness to cede control to another human and compare it to the willingness to cede control to an algorithm.[12] We plan to investigate this matter in future work by using data from the present study as the performance of another human in the task.

[12] In a similar vein, Dietvorst et al. (2015) and Logg et al. (2019) identified the role of authority when comparing human to algorithm choices. They report that algorithm uptake is lower in the choice between oneself and an algorithm compared to the choice between another human and an algorithm.

CRediT authorship contribution statement

Radosveta Ivanova-Stenzel: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Michel Tolksdorf:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Data availability

Data will be made available on request.

Appendix A

A.1. Additional figures

See Figs. A.1 and A.2.

A.2. Additional tables

See Tables A.1–A.3.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.socec.2024.102270.

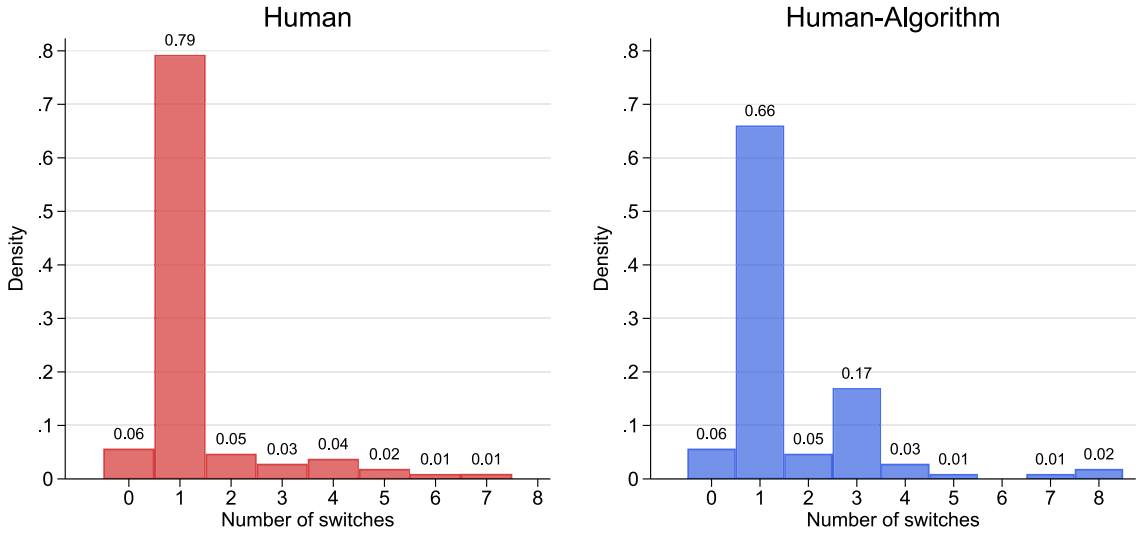


Fig. A.1. Relative share of number of switches between oneself and the algorithm in the choice problems by condition.

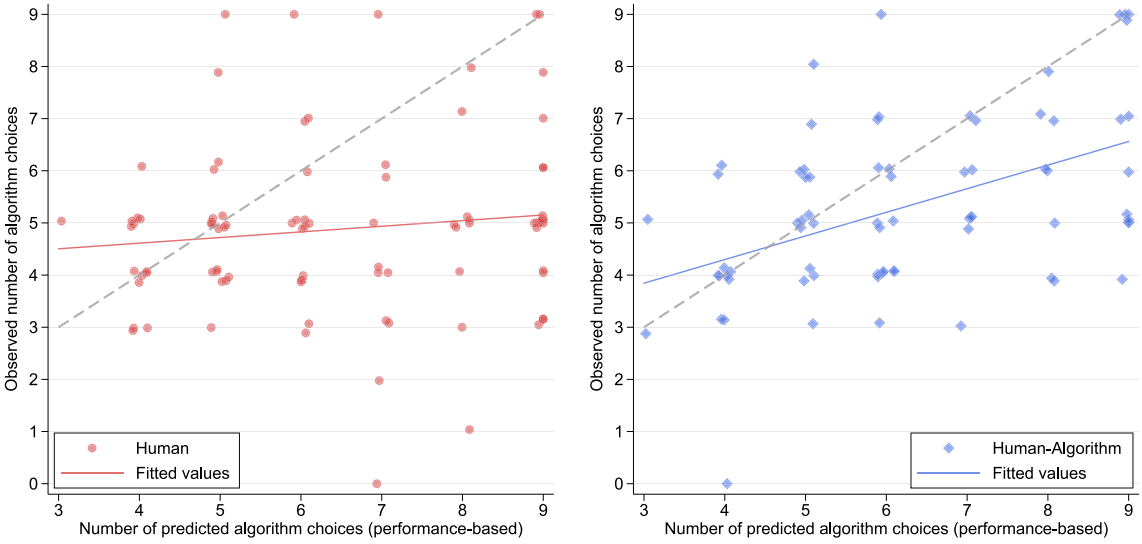


Fig. A.2. Observed and performance-based predicted number of algorithm choices per condition with fitted values (solid) and 45° line (dashed).

Table A.1
Belief and confidence questions.

How many ranks do you think the algorithm's estimate is away from the state's true rank? (0–50)
How many ranks do you think your estimate is away from the state's true rank? (0–50)
How much confidence do you have in your estimate? (1 = *none*; 5 = *a lot*)
How much confidence do you have in the algorithm's estimate? (1 = *none*; 5 = *a lot*)
How likely is it that you predicted the state's rank almost perfectly? (1 = *extremely unlikely*; 8 = *extremely likely*)
How likely is it that the algorithm predicted the state's rank almost perfectly? (1 = *extremely unlikely*; 8 = *extremely likely*)

Table A.2
Mean forecasting performance of participants and algorithm by condition with std. dev. in brackets.

	|own forecast – true rank|	|algorithm forecast – true rank|	Paired t-test
Unincentivized forecasts			
Human	6.61 (2.63)	4.3 (1.12)	$t(953) = 25.40,\ p < 0.01$
Human-Algorithm	6.88 (3.32)	4.36 (1.18)	$t(953) = 23.42,\ p < 0.01$
Incentivized forecast			
Human	6.26 (5.11)	3.84 (3.77)	$t(105) = 4.51,\ p < 0.01$
Human-Algorithm	7.08 (5.48)	3.74 (3.62)	$t(105) = 7.03,\ p < 0.01$

Table A.3
Interval regressions within and between conditions.

	Within conditions				Between conditions	
	Human		Human-Algorithm		Reference category: Human	
	(1)	(2)	(3)	(4)	(5)	(6)
Performance-based prediction	−0.044	−0.003	0.353	0.070	0.132	0.065
	(0.197)	(0.203)	(0.222)	(0.276)	(0.153)	(0.161)
Human-Algorithm					1.434***	1.410**
					(0.526)	(0.568)
Confidence in						
Confidence in forecast (own)	−1.132**	−1.124*	−0.518	−0.555	−0.875**	−0.923*
	(0.541)	(0.580)	(0.651)	(0.776)	(0.428)	(0.477)
Confidence in forecast (algorithm)	1.787***	1.974***	1.118	0.863	1.522***	1.604***
	(0.521)	(0.557)	(0.838)	(1.001)	(0.458)	(0.500)
Belief of						
Belief of deviation (own)	0.066	0.056	0.167	0.596**	0.086	0.129
	(0.091)	(0.092)	(0.159)	(0.248)	(0.084)	(0.096)
Belief of deviation (algorithm)	−0.036	−0.005	−0.119	−0.682**	−0.014	−0.055
	(0.130)	(0.133)	(0.195)	(0.298)	(0.111)	(0.123)
Perfect prediction (own)	−0.670	−0.708	−0.782	−0.411	−0.872**	−0.868**
	(0.464)	(0.514)	(0.523)	(0.512)	(0.356)	(0.390)
Perfect prediction (algorithm)	0.425	0.312	1.366***	1.075**	1.106***	1.063***
	(0.574)	(0.660)	(0.425)	(0.450)	(0.341)	(0.388)
Constant	13.276***	13.050***	6.338*	7.868**	7.234**	7.340**
	(3.494)	(3.542)	(3.756)	(3.617)	(2.860)	(2.864)
lnsigma	1.176***	1.189***	1.153***	1.144***	1.187***	1.201***
	(0.090)	(0.091)	(0.085)	(0.089)	(0.059)	(0.060)
Gender	Yes	Yes	Yes	Yes	Yes	Yes
Ambiguity and risk aversion	Yes	Yes	Yes	Yes	Yes	Yes
Only monotonic	No	Yes	No	Yes	No	Yes
Observations	100	85	98	68	198	153

Standard errors in parentheses. Estimation by interval regression. The dependent variable is the interval of penalty rate (X). The lower (upper) bound of the interval is determined by the first (last) switching point. For example, for a participant with three switches who switches from algorithm to own forecast in choice problem 3, then back to algorithm in choice problem 4, and then back to own forecast in choice problem 5, the interval of X is $[0.04, 0.12]$. Effect sizes are expressed in euro cents.

* Significance at the 10% level.

** Significance at the 5% level.

*** Significance at the 1% level.

Table A.4
OLS regressions within and between conditions (binary classification and fifth choice).

	Dependent variable: Binary classification			Dependent variable: Choice in problem 5		
	Human	Human-Algorithm	Reference category: Human	Human (4)	Human-Algorithm	Reference category: Human
	(1)	(2)	(3)		(5)	(6)
Performance-based prediction	0.106	0.432***	0.255***	0.022	0.366***	0.174**
	(0.127)	(0.117)	(0.090)	(0.121)	(0.121)	(0.086)
Human-Algorithm			0.048			0.005
			(0.062)			(0.065)
Confidence in						
Confidence in forecast (own)	−0.090	−0.165*	−0.095*	−0.042	−0.066	−0.043
	(0.064)	(0.090)	(0.052)	(0.064)	(0.101)	(0.052)
Confidence in forecast (algorithm)	0.129*	0.172**	0.132**	0.152**	0.139*	0.136***
	(0.074)	(0.083)	(0.054)	(0.065)	(0.080)	(0.049)
Belief of						
Belief of deviation (own)	0.009	0.003	0.006	0.014	0.007	0.010*
	(0.013)	(0.006)	(0.005)	(0.014)	(0.007)	(0.006)
Belief of deviation (algorithm)	−0.014	0.018	0.001	−0.003	−0.007	−0.002
	(0.017)	(0.018)	(0.011)	(0.017)	(0.020)	(0.010)
Perfect prediction (own)	−0.055	0.010	−0.056	−0.097	−0.028	−0.078*
	(0.061)	(0.066)	(0.043)	(0.061)	(0.071)	(0.043)
Perfect prediction (algorithm)	0.042	0.045	0.076	0.089	0.052	0.085*
	(0.083)	(0.057)	(0.047)	(0.085)	(0.056)	(0.047)
Constant	0.774*	0.025	0.314	0.832*	0.175	0.472
	(0.400)	(0.325)	(0.261)	(0.443)	(0.355)	(0.291)
Gender	Yes	Yes	Yes	Yes	Yes	Yes
Ambiguity and risk aversion	Yes	Yes	Yes	Yes	Yes	Yes
Within/Between conditions	Within	Within	Between	Within	Within	Between
Observations	106	106	212	106	106	212
R^2	0.226	0.291	0.224	0.236	0.216	0.199

Standard errors in parentheses. Estimation by OLS regression. The dependent variable is the binary classification of subjects' choices of the algorithm (0 corresponds to choosing the algorithm at most 4 times, 1 corresponds to choosing the algorithm at least 5 times) in specifications (1) to (3) and the subject's choice of the algorithm in the fifth choice problem in specification (4) to (6).

* Significance at the 10% level.

** Significance at the 5% level.

*** Significance at the 1% level.

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